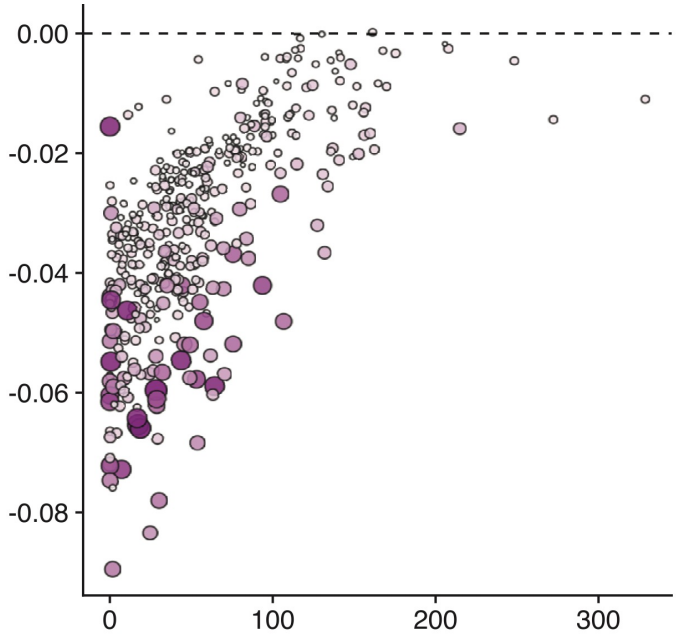
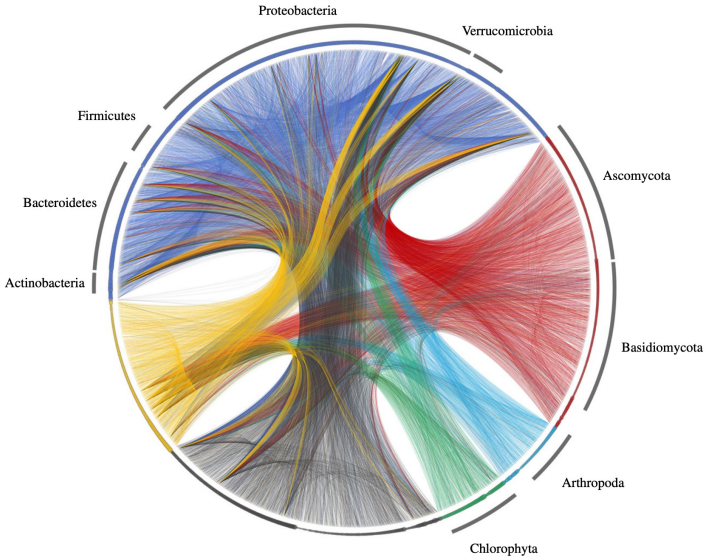
DASC 3240 Data visualization in Science





2023/24 SPRING SEMESTER

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Today's outline

1. What is "Data Visualization"?
2. Course information
3. (+ Basics in figure and data presentation)

What is “Data Visualization”?

Data Visualization

- "*Data visualization is the art and science of telling captivating stories with data*." (Belorkar et al. 2022)
- Scientists have to communicate insights effectively using data visualization (Belorkar et al. 2022).
- Data visualization is the final step of the process of **collecting**, **analyzing**, and **summarizing data**. Its purpose is to extract and summarize essential information about the data and effectively convey a key message to the audience.
- A good visualization is necessary for good science. A bad visualization fails to convey a key message, or may even lead to a **misunderstanding** of the data. Conversely, understanding data visualization is also necessary to detect misuse and misinterpretation of data.

Intended Learning Outcomes (ILOs)

- Elaborate the basic concept of visualization tools.
- Do simple data processing given a variaty of data formats with R/Python.
- Foumulate a visualization solution to some real-data problems and interpret the results by graphs.
- Implement some visualization techniques with R or Python.
- Apply the conceptual and practical skills to interpret data in physics, chemistry, life science and ocean science as well as other desciplines with data.

Course information

Course assessment

- In-Class Assignment: 40%
- Final projects: 60%

Course structure

- 1.5h lecture × 2 in each week:
  - 1st lecture in each week: Lecture
  - 2nd lecture in each week: Mostly hands-on + In-Class Assignment
  - **Students will be asked to submit the assignment during each class**

- 1h tutorial in each week:
  - Q&A, trouble shooting, face-to-face discussion, etc.
  - Students who need technical help and/or want to learn more about data visualization may attend the tutorial sesseions

Lecture schedule (Mon/Wed 12:00–13:20, Rm 4503)

Wk	L#	Date	Topic	Format	Assignments
1	1	31 Jan (Wed)	Course introduction Basics in figure and data presentation I	Lecture	
2	2	5 Feb (Mon)	Basics in figure and data presentation II	Lecture	
	3	7 Feb (Wed)	Basics in figure and data presentation III	Lecture	
3		12 Feb (Mon)	Lunar New Year	No class	
	4	14 Feb (Wed)	Review of Python basics	Lecture + hands-on	
4	5	19 Feb (Mon)	Overview of data visualization tools Plotting one variable: Density and histogram	Lecture + hands-on	
	6	21 Feb (Wed)		Hands-on	Yes
5	7	26 Feb (Mon)	Plotting two variables: Boxplot and violin plot	Lecture + hands-on	
	8	28 Feb (Wed)		Hands-on	Yes
6	9	4 Mar (Mon)	Plotting two variables: Scattered plot	Lecture + hands-on	
	10	6 Mar (Wed)		Hands-on	Yes
7	11	11 Mar (Mon)	Plotting two variables: Time series data	Lecture + hands-on	
	12	13 Mar (Wed)		Hands-on	Yes

Lecture schedule (Mon/Wed 12:00–13:20, Rm 4503)

8	13	18 Mar (Mon)	More plots: Dendrogram, Network and Map	Lecture + hands-on	
	14	20 Mar (Wed)		Hands-on	Yes
9	15	25 Mar (Mon)	Visualization of high dimensional data: PCA and others	Lecture + hands-on	
	16	27 Mar (Wed)		Hands-on	Yes
10		1 Apr (Mon)	Mid-term break	No class	
		3 Apr (Wed)	Mid-term break	No class	
11	17	8 Apr (Mon)	R and RStudio	Lecture + hands-on	
	18	10 Apr (Wed)	R basics – Introduction to R	Lecture + hands-on	
12	19	15 Apr (Mon)	R basics – Starting with data	Lecture + hands-on	
	20	17 Apr (Wed)	Data visualization with R and ggplot2 I	Lecture + hands-on	Yes
13	21	22 Apr (Mon)	Data visualization with R and ggplot2 II	Lecture + hands-on	Yes
	22	24 Apr (Wed)	Data visualization with R and plotly	Lecture + hands-on	
14	23	29 Apr (Mon)	Integrating R and other tools	Lecture + hands-on	
		1 May (Wed)	Labor Day	No Class	
15	24	6 May (Mon)	Final project		
	25	8 May (Wed)	Final project		Yes

Tutorial (Wed 18:00–18:50, Rm 2463)

Wk	T#	Date	Topic	Activity
1	1	31 Jan (Wed)	Setting up Python environment	Technical support for the setup
2	2	7 Feb (Wed)		TBA
3	3	14 Feb (Wed)	Review of Python basics	Technical support for Jupyter notebook
4	4	21 Feb (Wed)	Plotting one variable: Density and histogram	Technical support and programming tips
5	5	28 Feb (Wed)	Plotting two variables: Boxplot and violin plot	Technical support and programming tips
6	6	6 Mar (Wed)	Plotting two variables: Scattered plot	Technical support and programming tips
7	7	13 Mar (Wed)	Plotting two variables: Time series data	Technical support and programming tips
8	8	20 Mar (Wed)	More plots: Dendrogram, Network and Map	Technical support and programming tips
9	9	27 Mar (Wed)	Visualization of high dimensional data: PCA and others	Technical support and programming tips
10		3 Apr (Wed)	Mid-term break	No class
11	10	10 Apr (Wed)	Setting up R and RStudio environment	Technical support for the setup
12	11	17 Apr (Wed)	Data visualization with R and ggplot2	Technical support and programming tips
13	12	24 Apr (Wed)	Integration of R, Git, RStudio, and Python	Technical support for the setup
14		1 May (Wed)	Labor Day	No Class
15	13	8 May (Wed)	Final project	

Course materials

me

you

Presentation slides

Python codes
Other text resources

upload

Prepared with
jupyter notebook

upload

download is
not necessary

download

GitHub

https://github.com/ong8181/DASC3240

Option 1:
If you do not have Python and jupyter in your PC, you can execute the codes using Google Colab.

colab

Option 2:
If you have Python and jupyter in your PC, you can download the codes and execute them.

If you have a trouble(s), you can utilize a face-to-face assistant opportunity in the tutorial.